MMCS Group Project · 2026 · Tier 1

Recovering the value of surplus renewable generation

Battery or electrolyser, at what scale, and under which contract

PREPARED BY	VERSION	DATA WINDOW	DECISION OWNER
Group 15	**1.0 — 5 November 2026**	**1 Jan 2020 – 17 Sep 2026**	**Portfolio board**

HEADLINE RESULTS

+£8.57m	**20 MW**	**2.6 yr**	**0.07%**
annual value of the recommended 20 MW module	electrolyser; no battery is economic	payback before stack replacement	of current sales diverted to make hydrogen

Decision

[*Build / do not build — one sentence. Then: what it costs, what it earns, what it pays back, and the single condition that would reverse the answer.*]

1 Executive summary

[*First paragraph contains numbers, not framing. Capacity, annual value, payback, and the assumption that carries the answer. Two sentences of context at most.*]

[*Second paragraph: the mechanism. Why this configuration and not the alternative, in one or two sentences, naming the constraint that actually binds (a contract cap, a cost threshold, an operating rule) rather than the one a reader would guess.*]

[*Third paragraph: what the reader must accept. Data vintage, time resolution, foresight, and the one number that would move most if that assumption failed.*]

The assumption that carries the answer

[*Name it and quantify it. "The result holds while the offtake contract is at least X; at Y the investment is zero." A caveat box that cannot be falsified by a number is decoration.*]

2 Problem, boundary and data

2.1 The question

[*State the decision in one sentence, then the arithmetic that frames it: how much energy the system produces, how much demand it serves, how much is stranded, and on how many days. These four numbers justify the whole report.*]

2.2 System boundary

[*What is inside the model and what is not. Be explicit that constructed supply is not observed curtailment, and that the demand series is the absorption cap. One sentence per excluded mechanism (network, intraday, ancillary, storage of hydrogen), because each exclusion is later a limitation.*]

Figure slot 1 — system boundary diagram, 16.4 cm × 4.6 cm
generation → {sale, battery, electrolyser, unused} → demand

2.3 Data and provenance

Table 1. Data series, provenance and use.

Series	Period	Treatment	Where used
[*Generation by fuel*]	[*2020–2026*]	[*repaired release v2; 523 of 2,452 days carry a non-strict flag*]	[*all cases*]
[*Demand*]	[*2020–2026*]	[*truncated last day of each query restored*]	[*absorption cap*]
[*Market index price*]	[*2020–2026*]	[*half-scale placeholder providers removed*]	[*revenue*]

Source: [*repository path and release tag*]. Repair log: [*file*].

2.4 Conventions and assumptions

Table 2. Assumption register. "Tested in" points at the sensitivity that would expose the assumption.

Assumption	Value used	Source	Tested in
[*Discount rate*]	[*5% real*]	[*course brief, §2.3*]	[*§5.1*]
[*Hydrogen reference price*]	[*£7.80/kg*]	[*policy strike, deflated to 2020 money*]	[*§5.1*]
[*Battery round-trip efficiency*]	[*90%*]	[*brief; 20–80% usable window*]	[*§5.2*]

Money is constant [*2020*] GBP unless stated. Energy figures are daily averages of metered half-hourly data.

3 Model

3.1 Decision variables

[*One sentence per variable group, or a compact list. A reader must be able to reconstruct the model from §3.1–§3.3 without the code.*]

3.2 Objective

$$\max \ \Pi = \frac{1}{Y} \sum_{t} \left[\frac{p_t \, (q_t + d_t)}{1000} + \frac{v \, h_t}{\eta} \right] - a_B \, n_B - a_H \, n_H \tag{1}$$

[*Define every symbol immediately below the equation, in running text, not in a symbol table the examiner has to hunt for. Say what the units are and which term carries the money. Refer back to it as* (1) *— a number that is never cited in the text is a number nobody reads. A useful test: give the magnitude of each term for the base case, so a reader can see which one dominates and which one is a rounding error.*]

[*If the report quotes equivalent annual values, show the annualisation rather than describing it:*]

$$a_j = F_j + \frac{i}{1-(1+i)^{-N_j}}\Big(K_j + \sum_{k \in R_j} \frac{C_{jk}}{(1+i)^k}\Big), \qquad j \in \{B, H\} \tag{2}$$

[*with the life, discount rate and renewal years named in the text, and at least one worked value — fixed cost plus capital plus renewals for the asset that carries the decision.*]

3.3 Constraints

$$q_t + c_t + h_t + u_t = G_t \qquad \text{(daily energy balance)} \tag{3}$$

$$q_t + d_t \le D_t \qquad \text{(demand cap)} \tag{4}$$

$$0.096\, z_t \le h_t \le 0.48\, z_t \le 0.48\, n_H \qquad \text{(module limits)} \tag{5}$$

$$h_t \le \eta\, L / 10^6 \qquad \text{(offtake cap)} \tag{6}$$

$$\textstyle\sum_t d_t = 0.9 \sum_t c_t \qquad \text{(annual battery balance)} \tag{7}$$

[*After the block, one sentence per constraint that needs justification, and one sentence naming the relaxation. If a constraint is an upper bound on value rather than a representation of physics, say so here, not in the limitations.*]

3.4 Dispatch policies

[*If the report compares an optimised allocation with a rule, define both as maths, not as prose. Policy A: economic allocation. Policy B: current generation serves current demand first, i.e.* $q_t = \min(G_t, D_t)$*. State which one the client is permitted to run — that is a commercial fact, not a modelling choice.*]

3.5 Instance size and solver evidence

Table 3. Solver evidence for one design year. Times are wall-clock on [*hardware*], default solver parameters except where noted.

Case	Rows	Cols	Objective	Time	Status
			£m/yr	s	
[*Baseline, policy A*]	2,556	2,192	17,139.3	0.05	optimal
[*20 MW electrolyser*]	2,556	2,192	17,147.9	0.06	optimal
[*50 t/day contract*]	N/A	N/A	N/A	N/A	not run (licence limit)

[*Solver and version, licence class, gap settings, hardware, and the one setting that changed the answer. Write N/A, never leave a cell blank.*]

4 Results

4.1 Validation ladder

Table 4. Validation ladder: identical KPIs across four solution methods.

Method	Value £m/yr	Unused GWh	Coverage ratio	What it shows
[*No investment*]	0	549.9	0.875	[*reference case*]
[*Naive rule, same asset*]	−1.065	549.9	0.875	[*a fixed rule destroys value*]
[*Optimised, same asset*]	8.572	536.0	0.874	[*value of dispatch*]
[*Imperfect foresight*]	N/A	N/A	N/A	[*outside Tier 1*]

Increments are measured against the matching no-investment case, not against each other.

4.2 Configuration comparison

Table 5. Annual increment by configuration, £m/yr, for the reference year.

Policy	None	Battery	Electrolyser	Both
[*Policy A: economic*]	0	0	**8.57**	8.57
[*Policy B: supply first*]	0	0	0	0

Zero means the asset is not selected, not that the asset earns nothing. Bold marks the recommended configuration.

4.3 Where the energy comes from

Figure slot 2 — energy attribution, 11 cm × 3.9 cm
two-panel: input split by source; renewable share on surplus days

Figure 1. [*Attribution of electrolyser input to avoided waste and diverted sale, with the daily lower and upper bound on the renewable share.*]
Source: [*results file, case id*].

4.4 Selection

[*Three short paragraphs, one per rejected alternative. For each: the quantity that decides it, the threshold at which it would be selected, and whether the threshold is inside the plausible range. If an alternative fails for a reason the model cannot see, say which model in Tier 2 will see it.*]

5 Sensitivity, break-even and switching points

5.1 One-at-a-time sensitivity

Table 6. Effect of single-parameter changes on the annual increment (£m/yr), capacity held at the base case unless stated.

Parameter	Setting tested	Increment	Capacity changes?
[*Electrolyser CAPEX*]	[*£0.82m/MW, legacy*]	9.41	no
[*Electrolyser CAPEX*]	[*£1.65m/MW, high*]	8.33	no
[*Discount rate*]	[*0% real*]	9.44	no
[*Discount rate*]	[*12% real*]	6.99	no
[*Hydrogen price*]	[*£6.00/kg*]	3.33	[*20 MW*]
[*Hydrogen price*]	[*£9.50/kg*]	13.88	[*60 MW*]

State the number of cases and how they were generated; a sensitivity table with six rows and no description of the sweep is still a guess.

5.2 Break-even and switching points

Table 7. Thresholds at which the recommendation changes.

Threshold	Value	Comment
[*Netback needed at 10 t/day*]	4.623	[*£/kg; below this, do not build*]
[*Offtake needed at £7.80/kg*]	2,230	[*kg/day*]
[*Battery break-even capital cost*]	6,990	[*£/MWh on the low-FOM table*]

5.3 Critical map

[*Two decision parameters on the axes, the decision boundary drawn, and the base case marked. One figure replaces a page of two-way sensitivity prose.*]

5.4 What would falsify this

[*Three to five bullets, each a measurable event: a contract term, a price, a cost, or an obligation. "If the buyer requires X, the investment is zero" is falsifiable; "results may vary" is not.*]

6 Recommendation and implementation

1. [*Do this now — with the number that justifies it.*]
2. [*Negotiate this first — the term that gates the investment.*]
3. [*Stage the expansion — what to build at which contract size.*]
4. [*Keep this alternative visible — the obligation under which the answer reverses.*]

7 Validation and limitations

7.1 Checks performed

[*What was independently checked, how many checks, and what a check means here (energy balance closure, cap compliance, recomputation of revenue from the printed plan). Then the reproduction statement: which reference values were reproduced, and the sentence that distinguishes arithmetic agreement from endorsement of shared assumptions.*]

7.2 Quantified concessions

Table 8. Approximations and their measured cost.

Simplification	Cost, measured	Where it bites
[*Daily time step*]	[*no intraday arbitrage; battery value overstated*]	[*§4.5*]
[*Annual battery balance*]	[*relaxation, so an upper bound on value*]	[*battery cases only*]
[*Perfect foresight*]	[*not quantified in Tier 1*]	[*all cases*]

7.3 Possible objections

> **Weakest link, volunteered**
>
> [*Name the parameter you trust least, say why it is still the right choice, and give the number that would change your mind.*]

Reproducibility appendix

[*Everything a second team needs to regenerate every number: data release tag, package version, solver version and licence class, entry-point commands in order, runtime, and the digest of the results file. A one-page appendix of plain commands is worth more than a page of prose about methodology.*]

Artefact	Command or digest
[*Prepare inputs*]	`[python src/prepare_data.py]`
[*Run all cases*]	`[python src/run_cases.py]`
[*Verify*]	`[python src/verify.py]`
[*Results digest*]	`[sha256: 0000...]`

References

[1] Group 15 data-repair release [*20261104-v2*], [*2,452 days*], accessed [*access date*]; the repair log travels with the release.

[2] MMCS 2026/27 project brief, [*pp. 2–6*], and the teacher clarifications: asset modules, 10 t/day offtake cap, 5% real discount rate, £9.5/kg benchmark (2023 money).

[3] Office for National Statistics, Consumer Prices Index including owner occupiers' housing costs, series D7BT, monthly: the deflator to 2020 GBP.

[4] Cost sources with vintage: BEIS, *Hydrogen Production Costs 2021* (2020 GBP); IEA, *Global Hydrogen Review 2025*; Danish Energy Agency, *Technology Data for Energy Storage*; ITM POSEIDON data sheet.

[5] DESNZ, *HAR1 successful projects*, December 2023: weighted strike price £241/MWh (£175/MWh in 2012 prices). [*Name the independent implementation used for the cross-check: repository path, commit, date.*]

Template guide — delete before submission

Page budget

Section	Pages	What it must contain
Title block, key figures, decision	1	four numbers and a reversible claim, for the whole submission
Tier 1: summary, boundary, data	1	numbers first, provenance table, assumption register
Tier 1: model	1	objective, constraints, domains and integrality, solver evidence
Tier 1: results	1	validation ladder, configuration matrix, attribution figure
Tier 1: sensitivity and recommendation	0.5	one-at-a-time, break-even, numbered actions with numbers attached
Tier 2: model and results	3	the chosen extension, in the same structure
Tier 2: sensitivity and recommendation	1	break-even, falsifiers, numbered actions
Joint recommendation	1	the two layers resolved into one decision
References	0.5	sourced, no agent names, no placeholders
Total	10	the whole technical report, illustrations and references included

The ten-page figure is the course limit for the final technical report (teacher, launch session). The Tier 1 rows come from the blueprint's 3.5-page target; the Tier 2 rows are a placeholder — how Tier 2 splits internally is the group's decision, not this template's. An appendix neither counts towards the limit nor gets marked, so nothing the reader has to see belongs there.

Conventions this style enforces

- Figures and tables are numbered and referenced in text; captions sit above tables and below figures and carry a source note.
- Units live in the column header, not in every cell; thousands separators and a fixed number of decimals per column; no vertical rules, three horizontal rules per table.
- A cell that was not measured reads `N/A`; a blank cell is a defect, and in a numeric column write `{N/A}`.
- Sensitivity, break-even and switching points get a section of their own, and at least one figure per table in the results section.

Common defects this skeleton is designed to prevent

1. Opening with background instead of numbers.
2. A recommendation with no threshold at which it reverses.
3. Assumptions in prose, untraceable to a source.
4. Solver claims with no instance size and no failed cells.
5. A limitation section that is a list of apologies rather than measured costs.
6. Money and physical quantities added inside one expression.